

NSCLC: Multi-classification of subtypes using Gene expression from RNA-sequence

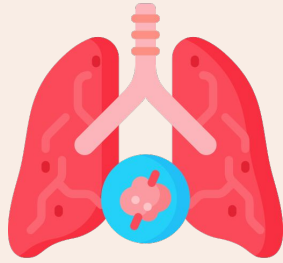
Kitiyaporn
Takham

(assisted by Jainam Modh)

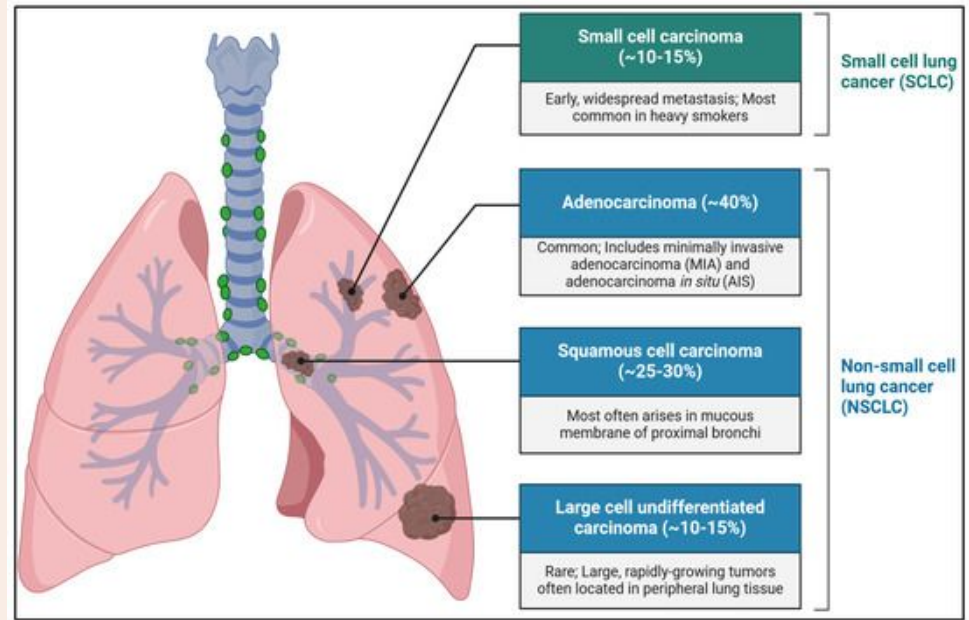


Background

Non-small cell Lung Cancer



85%

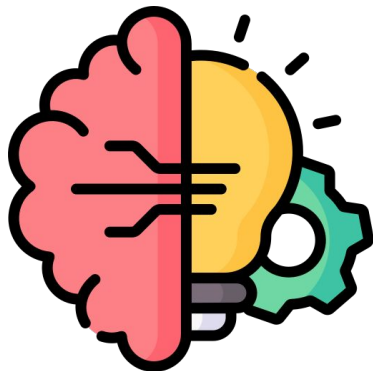


<https://www.prachachat.net/columns/news-1194487>

<https://www.nytimes.com/2022/11/06/health/to-bacco-fda-menthol-ban-nicotine.html>

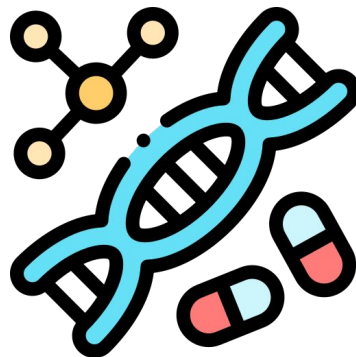
Garg et al., 2024

Statement of Modeling Task & Project Goals



Model Accuracy

- accuracy should be $\geq 80\%$
- F-1 score should be ≥ 0.80



Model Interpretation

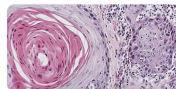
There is at least one gene expression that can be used for annotation for each subtype

Examples of Prior Modeling Work

Kaggle

TCGA - LUSC | Lung Cancer Gene Expression Dataset

LUNG CANCER - Lung Cell Squamous Carcinoma



Dataset

- **LUSC** gene expression profiling by array normalized by TPM
- 551 samples (cancer = 502, healthy = 49)
- 56,907 gene expression

 BRYAN BLAS RIMAC · 1Y AGO · 456 VIEWS

8

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⋮

TCGA_LUSC_PCA_CatBoost_BalancedAccuracy=100%

Model Approach 1

- Dimension reduction: PCA
- Model: CatBoostClassifier
- Matrices: Balanced Accuracy

Result: 100% Accuracy

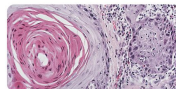
Bryam, 2024

Examples of Prior Modeling Work

Kaggle

TCGA - LUSC | Lung Cancer Gene Expression Dataset

LUNG CANCER - Lung Cell Squamous Carcinoma



Dataset

- **LUSC** gene expression profiling by array normalized by TPM
- 551 samples (cancer = 502, healthy = 49)
- 56,907 gene expression

BERKAY ÖZGELİK · 1Y AGO · 1,030 VIEWS

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Bioinformatics Approach | 99.1% Accuracy with DEGs

Model Approach 2

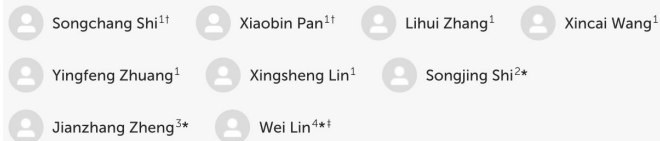
- Dimension reduction: DEGs
- Model: Logistic Regression, Support Vectors, Random Forest, CatBoostClassifier
- Matrices: Accuracy

Results: LR = 97.3%
 SVC = 97.3%
 RF = 96.4%
 CB = 99.1%

Examples of Prior Modeling Work



An application based on bioinformatics and machine learning for risk prediction of sepsis at first clinical presentation using transcriptomic data



Shi et al., 2022

Dataset

- **GSE185263** contains blood RNA-seq (transcriptomic data)
- 348 samples (348 = sepsis, 44 = healthy)

Model Approach

- Dimension reduction: RFE
- Model: Linear regression, Lasso regression, Decision Tree, Random Forest, CatBoostClassifier
- Matrices: Precision, Recall, f1-score

Results: combination of CatBoost and SHAP may serve as the best-performing machine learning model for predicting transcriptomic

	Scores		Test set					
			Precision		Recall		f1-scores	
	Train set	Test set	Normal	Sepsis	Normal	Sepsis	Normal	Sepsis
Linear	0.66	0.53						
LASSO Linear	0.66	0.53						
Logistic	0.99	0.94	0.69	0.97	0.75	0.96	0.72	0.97
Decision Tree	1.0	0.94	0.69	0.97	0.75	0.96	0.72	0.97
Random Forest	1.0	0.96	0.69	0.97	0.75	0.96	0.72	0.97
ANN-MLP	0.99	0.96	0.90	0.97	0.75	0.99	0.82	0.98
CatBoost	1.0	0.97	0.90	0.97	0.75	0.99	0.82	0.98

Summary of Dataset

GEO: GSE81089

- Gene expression of RNA-seq data in normalized **FPKM**
- Tumor tissue from **199 patients diagnosed with NSCLC** and 19 paired normal lung tissues
- surgically treated 2006-2010 at the Uppsala University Hospital, Uppsala, Sweden

Dataset

- 199 rows (patients)
- 63130 columns (genes)

Ensembl_gene_id	L400T	L401T	L404T	L406T	L413T	L414T	L417T	L420T
ENSG000000000003	52.195	37.8891	23.191	25.0324	41.9686	28.5794	19.9008	23.2632
ENSG000000000005	0.230061	0.0860342	0.0480224	0	2.5709	0.0871915	0.234047	0
ENSG000000000419	43.8616	47.0457	38.1292	54.303	51.2969	42.1604	78.1961	60.7283
ENSG000000000457	14.7101	7.81233	12.3117	8.41631	8.84999	6.28052	4.66235	7.39264
ENSG000000000460	4.81335	5.92073	8.21385	6.71221	4.79088	9.64764	6.60509	8.49746
ENSG000000000938	7.40831	9.83188	9.68575	10.9263	8.36149	1.38461	10.9656	10.8401
ENSG000000000971	112.426	39.7146	25.9596	80.2073	38.4429	87.5935	65.7782	27.1423
ENSG000000001036	43.9196	60.4056	49.0519	40.47	58.1048	67.6894	165.81	31.7792
ENSG000000001084	12.1289	9.20525	23.9222	46.9369	15.6082	22.3416	6.85872	289.794
ENSG000000001167	13.3027	19.6343	20.166	20.1807	23.3442	19.7386	15.8384	12.4058
ENSG000000001460	6.53823	2.65372	9.99002	5.55931	5.53239	7.13434	13.7378	5.83043
ENSG000000001461	11.9746	8.2977	14.6296	7.98718	8.30179	10.3533	9.83761	6.93396
ENSG000000001497	16.6233	15.7391	20.5829	14.8976	20.9113	12.4162	17.4688	15.6562
ENSG000000001561	19.1979	13.3403	30.3152	5.17111	34.8075	3.67391	16.123	4.43594
ENSG000000001617	5.9633	6.0369	5.54293	13.8534	3.81642	21.079	2.84586	13.9123
ENSG000000001626	3.37047	0.326229	1.01184	4.22917	19.523	8.15909	0.0111574	3.67477
ENSG000000001629	18.4514	21.8182	20.4802	17.0436	20.1822	21.43	46.9433	12.718
ENSG000000001630	35.4715	21.468	19.1304	22.218	43.2397	23.2196	75.7268	27.3399
ENSG000000001631	22.2211	16.5157	19.313	21.4983	23.3714	22.4684	38.918	17.2311
ENSG000000002016	4.71388	2.35516	1.90769	4.24452	3.58473	7.07046	2.53268	9.56792
ENSG000000002079	0.0156401	0	0.0229762	0.0649519	0.0144767	0.338066	5.54994	0.0448

• • •

Djureinovic D, Hallström BM, Horie M, Mattsson JS, La Fleur L, Fagerberg L, Brunnström H, Lindskog C, Madjar K, Rahnenführer J, Ekman S, Ståhle E, Koyi H, Brandén E, Edlund K, Hengstler JG, Lambe M, Saito A, Botling J, Pontén F, Uhlén M, Micke P

Modeling Approach

Data Preprocessing

1

Map gene id with human gene name, **Homo_sapiens.GRCh38.77**
Switch rows and columns

Ensembl_gene_id	L400T	L401T	L404T	L406T	L413T	L414T	L417T	L420T
ENSG000000000003	52.195	37.8891	23.191	25.0324	41.9686	28.5794	19.9008	23.2632
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ENSG000000001167	13.3027	19.6343	20.166	20.1807	23.3442	19.7386	15.8384	12.4058

Mapping

of gene names: 64253

	gene_id	gene_name
0	ENSG000000223972	DDX11L1
1	ENSG000000227232	WASH7P
2	ENSG000000278267	MIR6859-2
3	ENSG000000243485	MIR1302-10
4	ENSG000000274890	MIR1302-11

=

	L881T	L884T	L885T	L886T	L887T	L890T	gene_name
0	22.72840	26.425000	10.56880	29.84500	18.05750	44.87340	TSPAN6
1	0.00000	0.024609	0.00000	0.00000	0.00000	0.00000	TNMD
2	76.68320	53.666600	42.42780	65.96840	39.96450	38.39590	DPM1
3	4.02081	7.114000	7.67308	6.23066	6.56760	5.31170	SCYL3
4	9.37590	8.976050	2.83725	4.44909	6.03634	6.40973	C1orf112

Modeling Approach

Data Preprocessing

2 Concatenate gene expression with **subtype diagnosis** by `sample_id`

dataset shape: (199, 56688)

	sample_id	stage	diagnosis	TSPAN6	TNMD	DPM1	SCYL3	C1orf112	FGR	CFH	...
0	L400T	3	2	52.1950	0.230061	43.8616	14.71010	4.81335	7.40831	112.4260	...
1	L401T	5	2	37.8891	0.086034	47.0457	7.81233	5.92073	9.83188	39.7146	...
2	L404T	3	2	23.1910	0.048022	38.1292	12.31170	8.21385	9.68575	25.9596	...
3	L406T	1	1	25.0324	0.000000	54.3030	8.41631	6.71221	10.92630	80.2073	...
4	L413T	5	2	41.9686	2.570900	51.2969	8.84999	4.79088	8.36149	38.4429	...

Diagnosis class: 1=squamous cell cancer, 2=AC unspecified, 3=Large cell/ NOS

Modeling Approach

Data Preprocessing

3 Statistical analysis and data visualization

Split into 80 train : 20 test

	CFH	FUCA2	GCLC	NFYA	...
count	199.000000	199.000000	199.000000	199.000000	...
mean	62.071540	51.825785	55.933176	18.836817	...
std	35.885858	20.001301	76.434065	6.337184	...
min	4.284630	6.043070	2.781190	7.990450	...
25%	37.780450	38.714950	12.028800	14.093900	...
50%	53.892500	48.900400	25.282500	17.959000	...
75%	82.417200	60.553700	57.920100	21.792450	...
max	241.318000	165.810000	411.301000	44.409900	...

Class	Row count
1	67
2	108
3	24

Modeling Approach

CatBoost



CatBoost

Key Features

- CatBoostClassifier
- Feature Importance
- Model Visualization
- Tools Integration

CatBoost is a machine learning algorithm that uses **gradient boosting** on **decision trees**.

Ordered Boosting

Avoids target leakage and reduces overfitting by

Symmetric Tree Structure

Efficient training and GPU acceleration, fast inference, low variance and stable tree

Integration

Available for integrate in variety of tools such as Python, R

Modeling Approach



Imbalance handling

To reduce bias and make the model to learn equally



Feature reduction

Genomic data is high dimensional, need to reduce computational complexity



Grid Search + Cross-validation

Fine-tuning hyper-params to get the best model architecture fit to the dataset



Evaluation

- Balanced accuracy
(Sensitivity + Specificity)/2
- f1-score

Modeling Approach

Imbalance Handling

Setup experiments:

I. Class weight

Based on inverse frequency

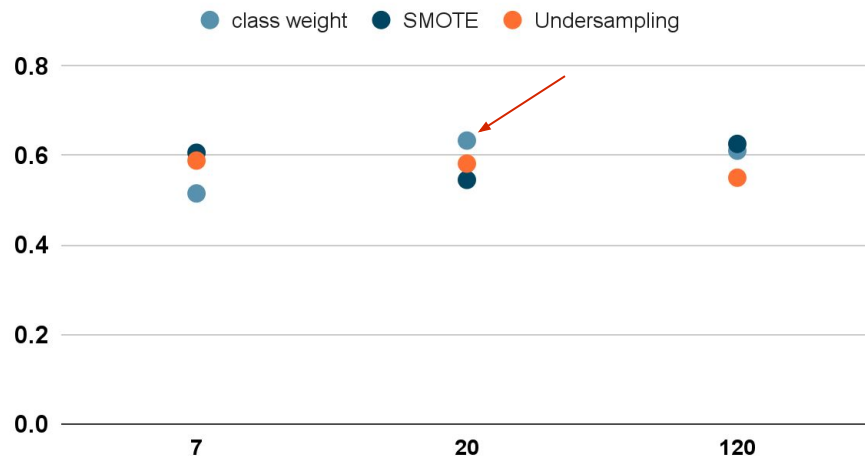
II. SMOTE

Generate synthetic data on the minor class based on neighborhood

III. Undersampling

Randomly pick from the majority class

Test Accuracy vs. #of component



The most appropriate method is **Class weight**

Modeling Approach

Feature Reduction

Setup experiments:

I. t-SNE

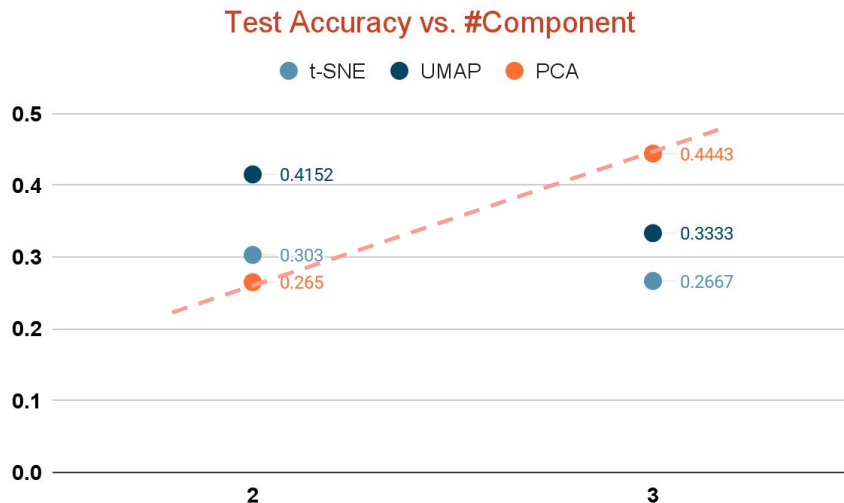
Non-linear correlation and reveal local structure
Limit to 2-3 dimension

II. UMAP

Non-linear correlation and reveal both local and global structures

III. PCA

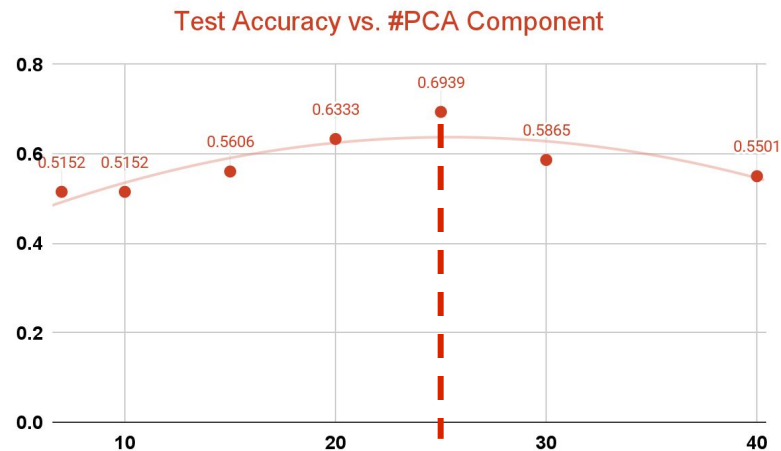
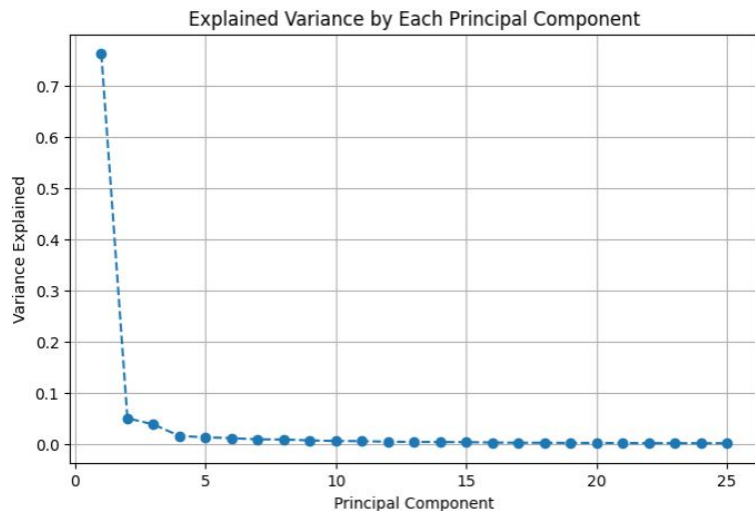
Linear correlation and reveal global structure



The most appropriate method is **PCA**

Modeling Approach

PCA Components



25 components gives the best performance

Modeling Approach



PCA

Components optimization
Result: **25 components** gave
the best performance



Grid Search + Cross-validation

Combine of hyperparameters
and train 5-fold cross validation
in **CatBoostClassifier** with
class weight



Evaluation

- Balanced accuracy
(Sensitivity + Specificity)/2
- f1-score



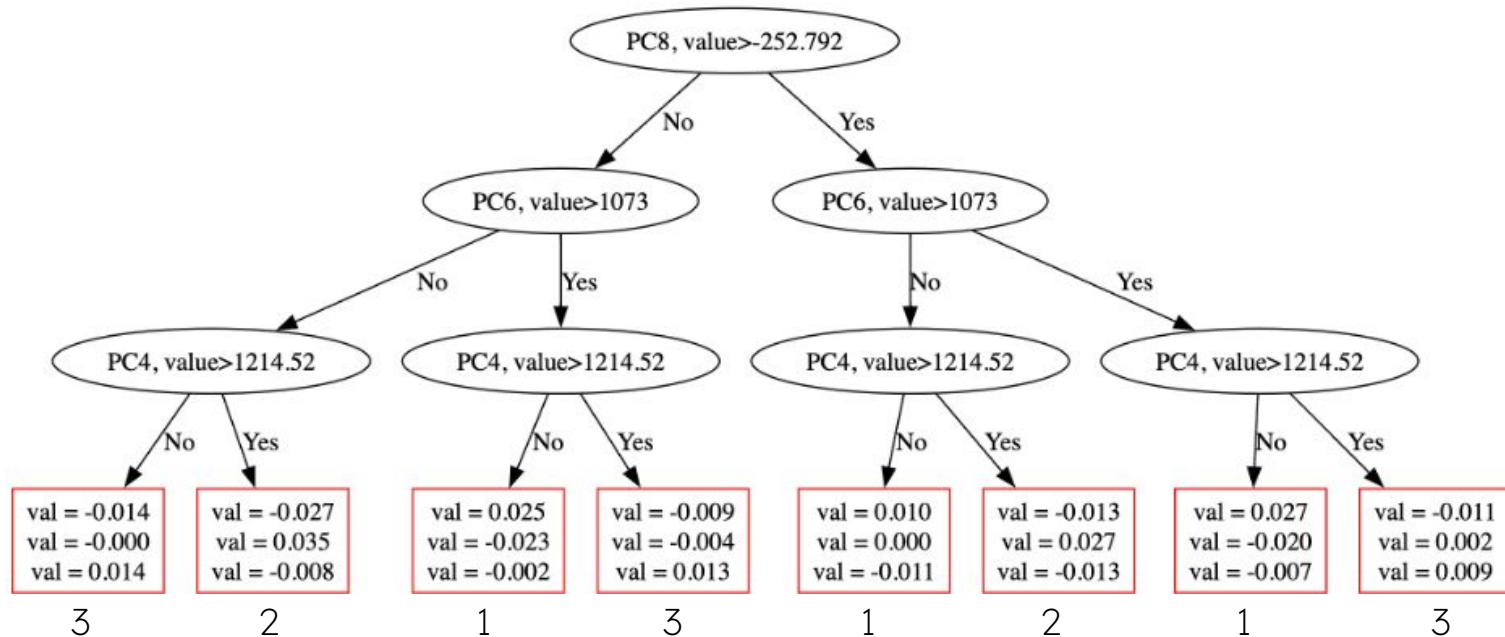
SHAP

Use for visualize importance
features based on Shapley
values to interpret the result

Summary of Result

Multi Classification model

Best Model: {'bootstrap_type': 'Bayesian', 'depth': 3, 'iterations': 300, 'l2_leaf_reg': 5, 'learning_rate': 0.05}



Results with Metrics

Model Comparison

Best Model Evaluation

Train Accuracy: 94.89%

Test Accuracy: 69.39%

F-1 score:

Squamous cell (0.812)

AC unspecified (0.789)

Large cell (0.400)



Model Accuracy

- accuracy should be $\geq 80\%$
- F-1 score should be ≥ 0.80

Project goal: **Failed** 😊

Confusion Matrix Train

	Squamous cell	AC unspecified	Large cell
Squamous cell	52	2	0
AC unspecified	9	76	1
Large cell	0	0	19

Predicted

Confusion Matrix Test

	Squamous cell	AC unspecified	Large cell
Squamous cell	13	0	0
AC unspecified	4	15	3
Large cell	2	1	2

Predicted

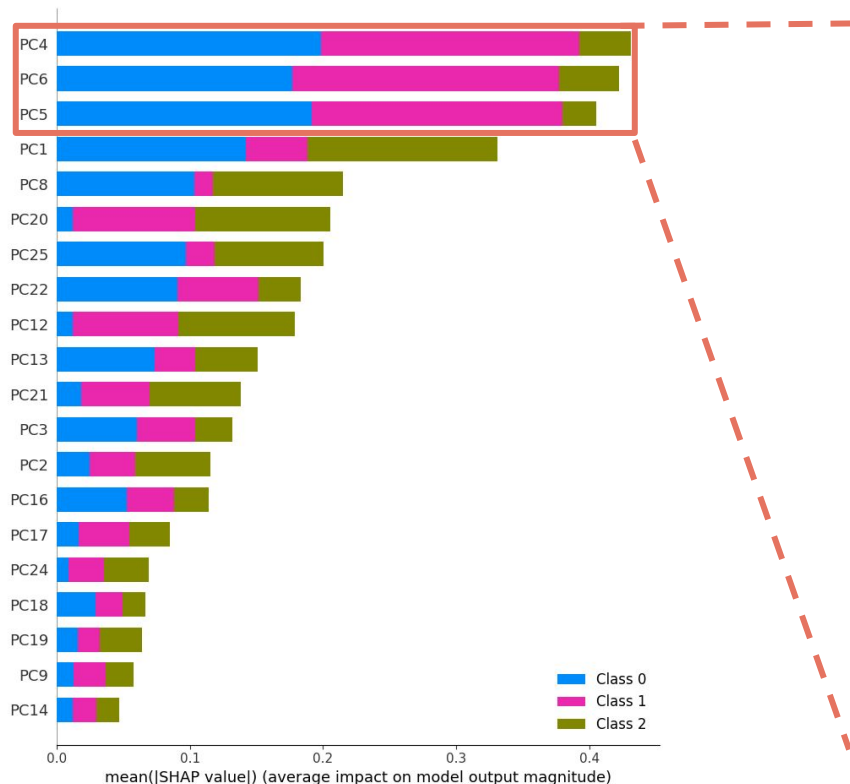
- 1 small and significantly imbalanced dataset
- 2 More complicated learning compared to others (binary classification)

Summary of Result

Model Interpretation



SHAP



Top 10 gene expressions

Class 1 (Squamous cell):

['SCYL3', 'C1orf112', 'FGR', 'TSPAN6', 'FUCA2', 'HS3ST1', 'BAD', 'LAS1L', 'DPM1', 'CFTR']

Class 2 (AC unspecified):

['FGR', 'SCYL3', 'C1orf112', 'RAD52', 'NIPAL3', 'BAD', 'MYH16', 'TSPAN6', 'DPM1', 'ANKIB1']

Class 3 (Large cell):

['TSPAN6', 'RAD52', 'FUCA2', 'NIPAL3', 'HS3ST1', 'MYH16', 'TNMD', 'LAS1L', 'FGR', 'SCYL3']

Summary of Result

Model Interpretation



Gene
Interpretation

Project goal:
Completed

CFTR

(Cystic Fibrosis Transmembrane Conductance Regulator)

An indicator for predicting adverse prognosis and metastasis in NSCLC patients

Li et al., 2015

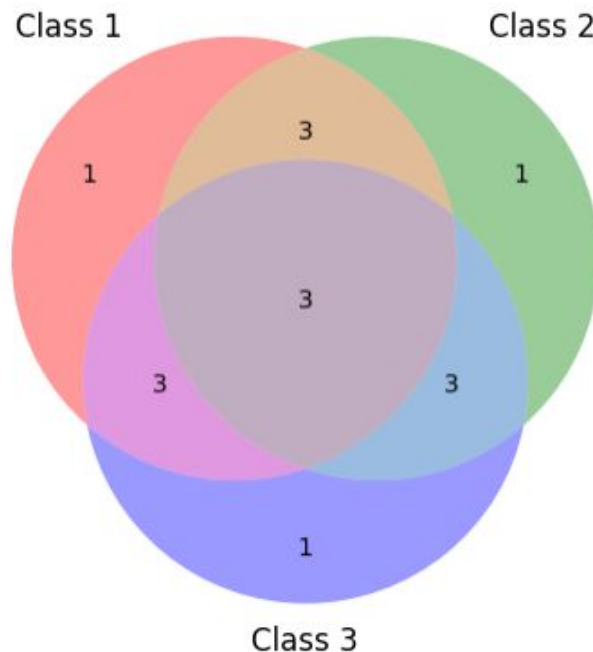
TNMD

(Tenomodulin)

Influence tumor-immune interactions and promote metastasis in various cancers

Chen et al., 2015

Overlap of Top Biomarker Genes Among Classes



ANKIB1

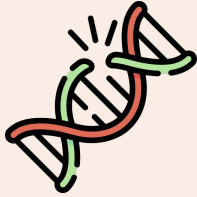
(Ankyrin Repeat and IBR Domain Containing 1)

One of the top 10 most significant correlations between RFTN1

Zhao et al., 2015

Diagnosis class: 1=squamous cell cancer, 2=AC unspecified, 3=Large cell/ NOS

Conclusion



Scientific

- CFTR may be used to identify squamous cell carcinoma
- TNMD may be used to identify large cell carcinoma

Note: Still need more study on identified gene



AI

- CatBoost library is a good option for decision tree model



Future Work

- Collect more data on Large cell carcinoma
- GeneXNet model (S. Vidhya and P. Velmurugan, 2024)
- Try more on feature extraction techniques such as Differentially Expressed Genes (DEGs) Analysis

References

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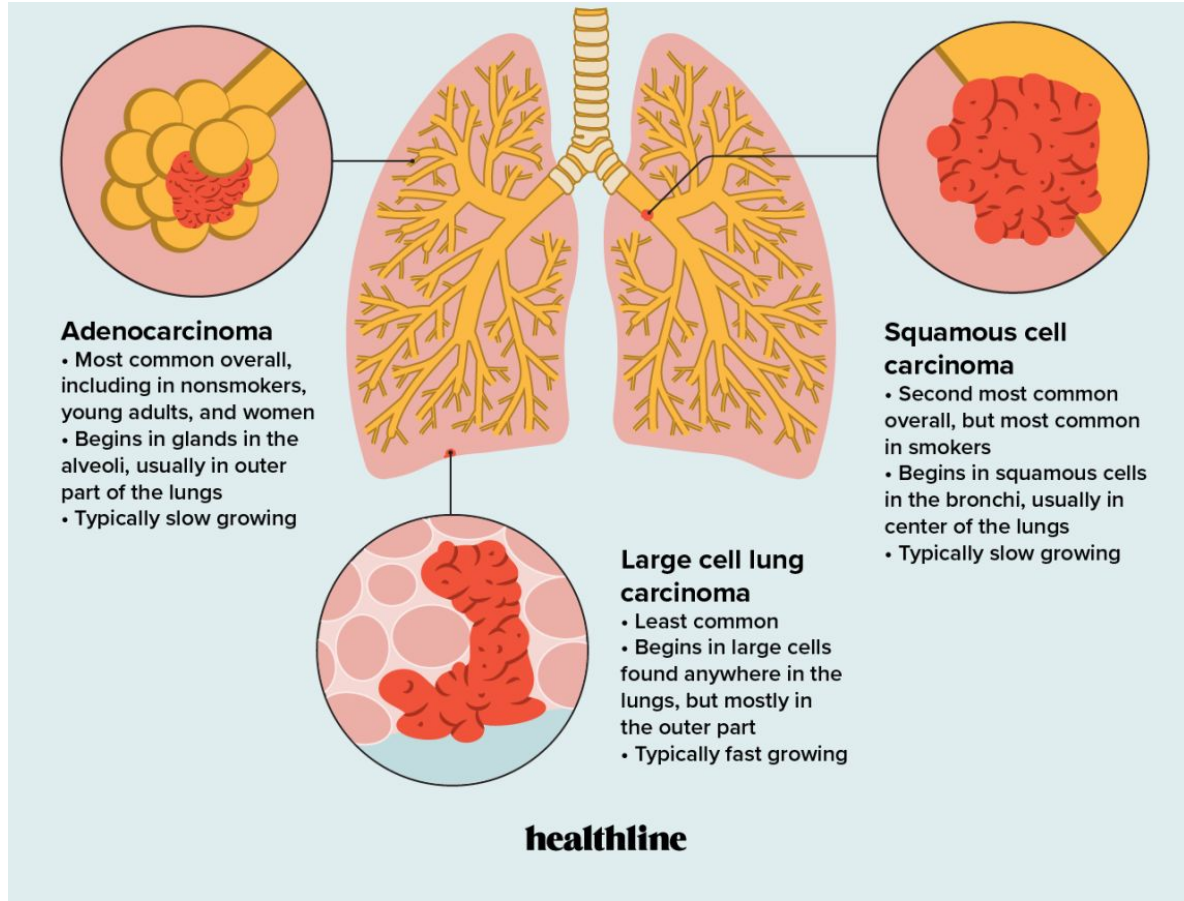
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Thank you

Any questions?

NSCLC Subtypes



RNA-seq data analysis: typical steps

STEP:

Quality control

Pre-processing

Alignment

Quality control

Quantitation

Combine count files to table

Quality control

Differential expression analysis

Fragments per Kilo Millions (FPKM)

Normalized read counts for

1. Sequencing depth
2. Length of the gene

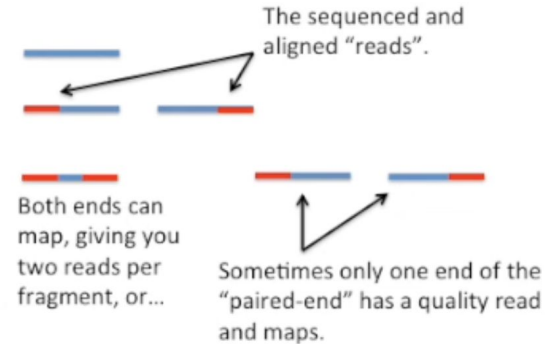
Considering for paired end of a fragment

A fragment to be sequenced:

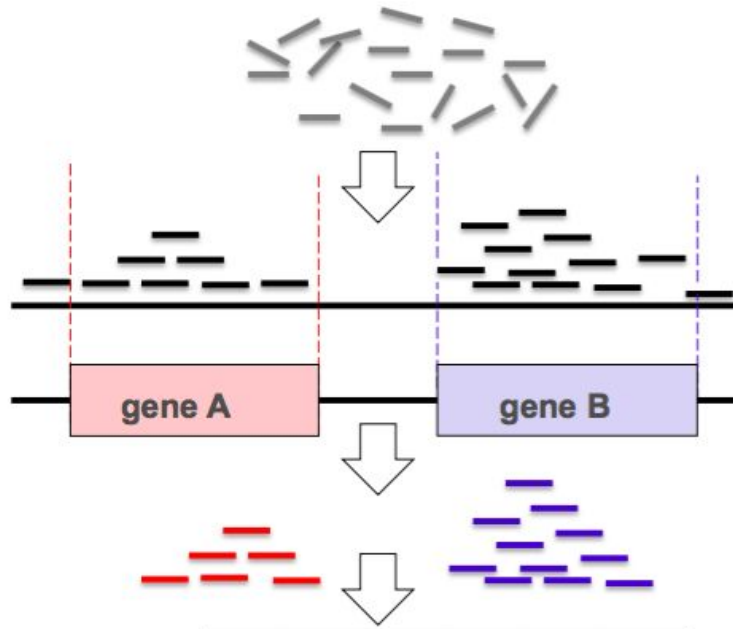
Single end sequencing:

Paired end sequencing:

FPKM keeps tracks of fragments so that one with two reads is not counted twice.



<https://www.youtube.com/watch?v=TTUrtCY2k-w>



	Control 1	Control 2	Control 3	Sample 1	Sample 2	Sample 3
Gene A	6	5	7	17	10	11
Gene B	11	11	10	3	4	2
Gene C	200	150	355	50	1	3
Gene D	0	1	0	2	0	1

Modeling Approach

Feature Reduction

Visual Summary

Step	PCA	t-SNE	UMAP
Assumes	Linear manifold	Nonlinear local neighborhoods	Nonlinear topological graph
Key algorithm	Eigen-decomposition	KL divergence minimization	Cross-entropy optimization
Outputs	Projection matrix (W)	Embedded points (no mapping)	Embedded points + graph
Generalizable	✅ Yes	❌ No	✅ Yes (partially)
Speed	⚡ Very Fast	🐢 Slow	⚡⚡ Fast

Modeling Approach

Feature Reduction

Setup experiments:

I. t-SNE

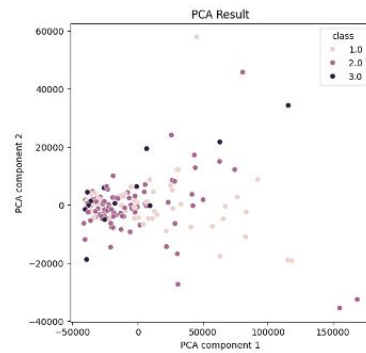
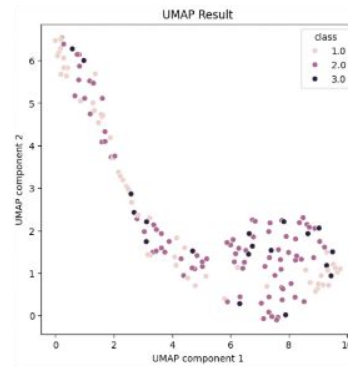
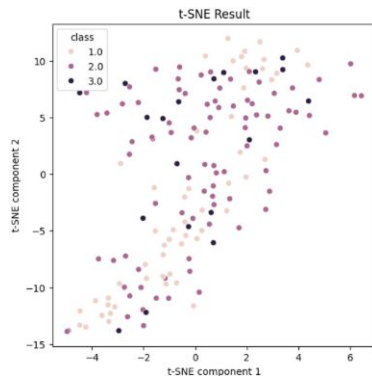
Non-linear correlation and reveal local structure
Limit to 2-3 dimension

II. UMAP

Non-linear correlation and reveal both local and global structures

III. PCA

Linear correlation and reveal global structure



SHAP Interpretation

$X =$	f1	f2	f3	f4

In this example, the training set contains "n" samples and 4 features

f1	f2	f3	f4

Training set

A random representative set of samples is taken from training set

f1	f2	f3	f4

Background set

Given an already trained ML model, the expected value $E[x]$ is obtained. In case of linear regression, $E[x]$ is the average of all predictions in the training set.

Figure 1 illustrates the proposed framework for multi-class classification. It shows three parallel processing paths for different coalitions: Coalition(i), Coalition(i+1), and Coalition(i+2).

Coalition(i) Path:

- Input: $z_{(i)}(1, 0, 1, 0)$ (Coalition(i))
- Feature Matrix:

f1	f2	f3	f4
x1	b2	x3	b4

 Brackets indicate: f1, f2, f3 are "Taken from the real 'x' instance"; f4 is "Taken from the background set".
- ML Model: The feature matrix is input to an "ML Model".
- Prediction: The model outputs $\bar{y}_{(i)}$ ("Prediction for coalition (i)").
- Weight: The prediction is multiplied by $w_{z(i)}$ ("Weight for coalition (i)").

Coalition(i+1) Path:

- Input: $z_{(i+1)}(1, 1, 1, 0)$ (Coalition(i+1))
- Feature Matrix:

f1	f2	f3	f4
x1	x2	x3	b4

 Brackets indicate: f1, f2, f3 are "Taken from the real 'x' instance"; f4 is "Taken from the background set".
- ML Model: The feature matrix is input to an "ML Model".
- Prediction: The model outputs $\bar{y}_{(i+1)}$ ("Prediction for coalition (i+1)").
- Weight: The prediction is multiplied by $w_{z(i+1)}$ ("Weight for coalition (i+1)").

Coalition(i+2) Path:

- Input: $z_{(i+2)}(1, 0, 0, 0)$ (Coalition(i+2))
- Feature Matrix:

f1	f2	f3	f4
x1	b2	b3	b4

 Brackets indicate: f1, f2 are "Taken from the real 'x' instance"; f3, f4 are "Taken from the background set".
- ML Model: The feature matrix is input to an "ML Model".
- Prediction: The model outputs $\bar{y}_{(i+2)}$ ("Prediction for coalition (i+2)").
- Weight: The prediction is multiplied by $w_{z(i+2)}$ ("Weight for coalition (i+2)").

Once optimized the weighted linear model, the coefficients are the Shapley values!

GeneXNet

Deep Learning for Multi-Tissue Cancer Classification of Gene Expressions (GeneXNet)

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